

End of the Road: The End of the Road

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15. *Staphylococcus aureus* is a Gram-positive, spherical bacterium that is commonly found on the skin and in the nasal cavity of humans. It is a facultative anaerobe, meaning it can grow in the presence or absence of oxygen. *S. aureus* is a major cause of skin infections, such as abscesses and boils, and is also responsible for more serious infections, including pneumonia and sepsis. It is highly resistant to many antibiotics, making it a significant public health concern.

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Let's suppose we have a function $f(x)$ defined on the interval $[a, b]$. We want to find the area under the curve $y = f(x)$ from $x = a$ to $x = b$. This is done by approximating the area with rectangles. The more rectangles we use, the better the approximation.

Let's take a specific example. Suppose $f(x) = x^2$ and we want to find the area under the curve from $x = 0$ to $x = 1$. We can use the formula for the area of a rectangle, $A = \text{width} \times \text{height}$, to approximate the area.

Let's divide the interval $[0, 1]$ into n subintervals. The width of each subinterval is $\Delta x = \frac{1}{n}$. The height of each rectangle is $f(x_i)$, where x_i is the right endpoint of the subinterval.

The area of each rectangle is $A_i = \Delta x \cdot f(x_i)$. The total area is the sum of the areas of all rectangles: $A \approx \sum_{i=1}^n \Delta x \cdot f(x_i)$.

As n increases, the approximation becomes more accurate. In the limit, as $n \rightarrow \infty$, the approximation becomes the exact area under the curve.

This process is called Riemann integration. The limit of the sum of the areas of the rectangles as $n \rightarrow \infty$ is the definite integral of $f(x)$ from a to b . The definite integral is denoted by $\int_a^b f(x) dx$.

The definite integral has many properties. One important property is that it is linear. If $f(x)$ and $g(x)$ are functions, then $\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$.

Another important property is that the definite integral is invariant under a change of variables. If $u = g(x)$, then $\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$.

The definite integral is also used to find the area of a region. If a region is bounded by a curve $y = f(x)$ and the x-axis, then the area of the region is $\int_a^b f(x) dx$.